LearnTrack: An Exploratory Data Analysis Dashboard for Student Performance Analytics

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***Abstract*—Academic records are often stored in tabular form, making it difficult for instructors to compare variables, inspect distributions, and communicate performance patterns quickly. This paper presents *LearnTrack*, a faculty-oriented Student Performance Analytics Dashboard that combines exploratory data analysis (EDA), descriptive statistics, Pearson correlation, linear regression summaries, and interactive visualizations. The analysis uses the supplied `clean_student_dataset.csv`, containing 500 records and 11 fields. The file has no missing values and no duplicate rows. In the sample, mean examination score is 69.79/100 (SD 12.83), mean attendance is 71.91%, and mean daily study time is 5.03 hours (SD 2.34). Daily study hours have a strong positive association with examination score ($r = 0.779$, 95% CI $[0.742, 0.811]$, $p < 0.001$); attendance also has a positive association ($r = 0.502$, 95% CI $[0.433, 0.565]$, $p < 0.001$). The association between study hours and attendance is small and not statistically significant ($r = -0.055$, $p = 0.219$). A React/Vite dashboard deployed through Vercel presents cohort summaries, distribution plots, correlations, course comparisons, and student-record views. The CSV-derived statistics differ slightly from some values rendered in the supplied dashboard screenshots; the paper reports independently recomputed CSV results and documents the discrepancies. Since the original dataset source and sampling procedure are unknown, findings are descriptive of this file and should not be generalized without independent validation.**

***Index Terms*—student performance, exploratory data analysis, learning analytics, attendance, study hours, correlation analysis, educational dashboard, data visualization**

I. Introduction

Student academic performance is influenced by a range of factors, and instructors may need to inspect attendance, study time, and examination scores together when reviewing a cohort. Reading raw tables manually can make distributions and relationships difficult to recognize, particularly when data spans multiple courses. Exploratory data analysis (EDA) provides an accessible starting point: it describes the data, exposes possible irregularities, and generates questions for further investigation without presuming a predictive model.

Educational data mining and learning analytics have developed methods for analyzing student records, learning activity, and educational outcomes. Surveys and reviews describe a broad range of prediction, clustering, visualization, and decision-support approaches [1]–[8]. Research on attendance has separately investigated its relationship with academic outcomes, with results depending on the setting, controls, and research design [9]–[14], while research on study behavior cautions that study time alone is an incomplete measure of learning [15]–[17]. Research on learning analytics dashboards examines how such information can be represented to support sense-making, feedback, and instructional decisions [18]–[30], and work on ethics and privacy stresses responsible use of student data [31]–[33].

This paper describes *LearnTrack*, a compact, interactive dashboard whose current analytical emphasis is descriptive rather than predictive. It integrates cohort-level metrics, charts of study time and attendance, performance distributions, course-level summaries, a correlation heatmap, and an interface for inspecting student rows. Its purpose is to make patterns easier to inspect, not to establish that a variable causes higher grades or to automate consequential decisions about students.

The contributions are: (1) a reproducible EDA workflow for a dataset containing attendance, study time, and examination scores; (2) CSV-derived descriptive and correlation results for 500 records; (3) a dashboard design that exposes results through multiple linked views; and (4) a focused comparison of 10 closely related research papers presented in one literature table, supported by a broader review of 45 references across educational data mining, attendance and study-time research, learning analytics dashboards, and responsible educational data use. The contribution is an applied system and analysis workflow; the paper does not claim that correlations or learning dashboards are novel in themselves.

II. Related Work and Literature Survey

The related literature is organized around five themes: (i) educational data mining and student-performance prediction, (ii) class attendance and study time in relation to academic achievement, (iii) learning analytics dashboards and visualization, (iv) early-alert systems and the learning analytics cycle, and (v) ethics and responsible use of student data. These themes are connected but should not be treated as interchangeable.

Educational data-mining papers commonly evaluate predictive models; attendance studies often use econometric or meta-analytic designs; dashboard papers examine how information is represented and used. LearnTrack is positioned at the intersection of these areas, with an emphasis on transparent exploratory data analysis rather than causal inference or prediction.

Educational data mining (EDM) and learning analytics research spans descriptive analysis, clustering, classification, early-warning systems, and predictive modeling. Early and updated surveys by Romero and Ventura, together with reviews by Dutt et al., Shahiri et al., Zhang et al., Roslan and Chen, and Alwarthan, summarize the methods, inputs, and evaluation choices used in this field [1]–[8]. Individual studies show how assignment marks, learning-management-system (LMS) activity, and ensemble models can be used for outcome prediction [34]–[37], and the Open University Learning Analytics Dataset provides a documented public resource for such work [38]. LearnTrack differs in scope: it reports descriptive statistics, Pearson correlations, and simple linear regression summaries, and it makes no claims about predictive accuracy or automated early warning.

Attendance research provides a rationale for examining attendance alongside grades, but the estimated relationship depends on the learning context, the variables controlled for, and the research design. Romer, Devadoss and Foltz, Stanca, and Latif and Miles examine attendance and performance using empirical designs ranging from cross-sectional regressions to panel data, and Credé et al. synthesize evidence across many samples [9]–[13]. Andrietti and Velasco consider attendance and study time together, which makes their work especially relevant to the variables selected for LearnTrack [14]. Study-behavior research adds a caution: Plant et al., Credé and Kuncel, and Nonis and Hudson indicate that hours spent studying do not capture study quality, skills, or habits [15]–[17]. Broader syntheses of achievement correlates, such as Richardson et al. and Schneider and Preckel, show that performance depends on many psychological and contextual variables beyond attendance and study time [39], [40]. The supplied dataset contains a daily-study-hours field but no validated measure of study quality, so study hours are treated as an observed variable only.

Dashboard research focuses on translating data into displays that educators and learners can interpret. Chatti et al. propose a reference model for learning analytics, Ferguson and Siemens discuss the field's drivers and emergence as a discipline, and Clow frames analytics as a cycle that should close the loop between data and action [18]–[20], [41]. Verbert et al. survey learning dashboard applications and open research directions, and Park and Jo describe the development of a dashboard for supporting learning performance [21]–[23]. Systematic reviews by Schwendimann et al., Matcha et al., Viberg et al., Verbert et al., and Paulsen and Lindsay identify gaps in theoretical grounding, evaluation methods, and evidence of learning impact [24], [25], [29], [42], [43]. Susnjak et al. discuss dashboards that provide actionable insights, Pei et al. present LearningViz for examining performance gaps, and Cerezo et al. discuss the convergence of EDM and learning analytics toward educational data science [26]–[28]. Cabral et al. review AI-powered dashboards and note limits in causal evidence, real-classroom deployment, and transparency [30]. Together these papers support combining cohort summaries, distributions, correlation views, and course comparisons, while indicating that a dashboard's existence is not proof of improved outcomes.

Early-alert and intervention systems connect analytics to action. Arnold and Pistilli describe Course Signals, and Jayaprakash et al. report an open-source early-alert initiative and its data-collection, portability, and evaluation issues [44], [45]. LearnTrack provides a cohort overview but has not evaluated any intervention. Finally, privacy and ethics research emphasizes responsible handling of learner records: Pardo and Siemens propose ethical and privacy principles, Ifenthaler and Schumacher study student perceptions of privacy, and Gašević et al. argue that learning analytics must remain about learning [31]–[33]. LearnTrack should therefore be used as decision support, with access control, data minimization, clear provenance, and human review.

Table I summarizes the 10 most closely related papers, selected to cover each theme; the remaining references cited in the text provide further context. It is a focused narrative review, not a systematic review, and does not claim exhaustive coverage of the field.

A. Synthesis and Research Gap

Attendance-focused studies and meta-analyses establish that attendance is a commonly examined academic factor, while warning that conclusions depend on study design and context [9], [11], [13], [14]. Study-behavior research adds an important qualification: reported study duration may not capture study quality or learning strategy [15], [17], and wider syntheses show that many other variables are associated with achievement [39], [40]. Accordingly, this paper interprets the observed associations as descriptive evidence rather than proof of a causal effect.

EDM reviews show extensive work on classification, prediction, and mining educational records [2], [4]–[7], and individual studies demonstrate prediction from marks and LMS logs [34], [35], [37]. LearnTrack is complementary rather than a replacement for these systems: its current claims concern distributions and pairwise linear associations, not predictive discrimination or intervention efficacy. Predictive accuracy that LearnTrack has not measured should not be quoted in comparison.

Dashboard research focuses not only on the calculation of metrics but also on whether users can interpret and act on them [22], [25], [27], [41]. Review work identifies gaps in theoretical grounding, evaluation methods, and evidence of learning impact [24], [29], [42], [43]. It is therefore important that LearnTrack's interface not be mistaken for evidence of improved learning: a future faculty study is needed to measure usability and decision quality. Early-alert studies likewise show that implementation and evaluation concerns must be addressed before analytics can

TABLE I
LITERATURE SURVEY: 10 CLOSELY RELATED RESEARCH PAPERS

Paper and year	Theme	Focus / method	Main contribution and relevance to LearnTrack
Shahiri et al. (2015) [4]	EDM / prediction	Review of data-mining methods for predicting student performance	Reviews prediction approaches and commonly used features. LearnTrack intentionally prioritizes transparent descriptive analysis over model-based prediction.
Romero and Ventura (2020) [5]	EDM / LA	Updated survey of EDM and learning analytics	Describes the relationship between EDM and learning analytics; informs positioning of LearnTrack as exploratory analytics rather than a prediction-only system.
Credé et al. (2010) [13]	Attendance	Meta-analysis of attendance, grades, and student characteristics	Synthesizes evidence across studies and motivates attendance as an academic variable. It provides broad motivation, not a row-level dashboard.
Andrietti and Velasco (2015) [14]	Attendance and study time	Panel-data study of lecture attendance, study time, and performance	Studies the same broad factors used by LearnTrack. Its econometric panel design differs from this paper's cross-sectional descriptive analysis.
Plant et al. (2005) [15]	Study time	Study time, study quality, and GPA	Argues that time quantity alone is an incomplete indicator; LearnTrack therefore presents study hours as an association, not a deterministic explanation of scores.
Verbert et al. (2014) [22]	Dashboards	Overview of learning dashboards and research opportunities	Surveys dashboard designs and open directions. LearnTrack applies this design tradition to a small set of academic measures.
Matcha et al. (2020) [25]	Dashboards	Systematic review from a self-regulated learning perspective	Identifies limitations in theoretical grounding and evaluation; LearnTrack should test interpretability with users rather than infer usability from implementation alone.
Pei et al. (2024) [27]	Dashboards	LearningViz: visual analysis of learning-performance gaps	Presents an interactive dashboard for student/group performance and item analysis. LearnTrack's scope is smaller and has no course-item analysis.
Cabral et al. (2025) [30]	AI dashboards	Systematic review of AI-powered learning analytics dashboards (21 studies, 2013–2024)	Identifies limits in causal evidence, classroom deployment, and transparency. Relevant to treating LearnTrack's AI-synthesis panel as optional and unevaluated.
Pardo and Siemens (2014) [31]	Ethics / privacy	Ethical and privacy principles for learning analytics	Supports minimization, transparency, and context-aware handling of learner records; informs LearnTrack's access-control and human-review requirements.

support intervention [44], [45], and privacy research supports minimization, transparency, and context-aware handling of learner records [31]–[33].

Recent work on AI-powered dashboards notes recurring limits in real-classroom deployment, causal evidence, and transparent use of AI outputs [30]. Two boundary decisions follow. First, LearnTrack is described as a faculty-oriented exploratory prototype, not as a demonstrated mechanism for improving grades. Second, the optional AI-synthesis panel is not part of the statistical method and its output is not counted as a validated result.

The project gap addressed here is practical integration: the dashboard places cohort descriptors, distributions, pairwise relationships, course summaries, and individual-row review in one compact interface. The gap should not be interpreted as an absence of prior dashboards or attendance analysis. The contribution is the applied workflow, the descriptive analysis of the supplied file, and explicit reporting of data provenance and output-consistency limitations.

Three design implications follow from the reviewed studies. First, attendance and study time are defensible variables for exploratory investigation, but neither should be treated as a standalone explanation of performance. Second, prediction-focused systems require careful validation and clearly defined target outcomes [36]. Third, visual summaries should expose both the direction and strength of associations and make the limitations of the underlying data visible. A next-stage evaluation should test whether instructors interpret these views correctly, whether the dashboard reduces time spent on routine analysis, and whether its insights lead to appropriate support actions, using user tasks, a usability instrument, and a pre-specified protocol.

The current evidence base for this implementation also has limitations: the file's sampling frame and source are unknown, the operational definition of study hours is not documented, and the supplied dashboard screenshots show some values that differ from recomputed CSV statistics. Unlike documented public resources such as the Open University dataset [38], the provenance of the supplied CSV is not yet established. Until these issues are resolved, the most defensible claim is that LearnTrack is a working exploratory analytics prototype that summarizes one supplied dataset, not a validated intervention or a generalizable student-performance model.

III. CONCEPTUAL FRAMEWORK AND RESEARCH QUESTIONS

A. From Raw Records to Interpretable Evidence

The project is organized around a simple evidence chain. Raw records are first treated as data that require inspection, not as facts that are automatically trustworthy. Column definitions and ranges are reviewed, missingness and duplicated rows are checked, and only then are summary measures calculated. Visual encodings are used to expose distributions and relationships that may not be obvious in a table. Finally, the presentation layer allows a user to move between a cohort overview, focused analytical views, and individual records where authorized. Each step has a different purpose: quality control establishes what can be computed; statistics summarize the file; visualizations support human inspection; and interface navigation supports workflow. None of these steps independently establishes causality or confirms that an intervention will improve learning.

This evidence chain is important in education because a dashboard can look authoritative even when the underlying fields have incomplete definitions. A percentage may be based on a threshold chosen by a developer, a category may have been computed from the same score displayed beside it, and a trend may reflect the way records were generated rather than a real educational process. The present study therefore treats the supplied CSV as the quantitative source of record

for every reported statistic and uses dashboard screenshots as evidence of the interface only. Where a screenshot disagrees with a recomputed result, the difference is reported rather than silently reconciled.

B. Scope of the Analytic Framework

LearnTrack combines descriptive analytics and bivariate association analysis. Descriptive analytics answers questions such as how scores are distributed, how many records fall into each supplied category, and how means differ across the course labels. Association analysis asks whether two numeric variables tend to vary together in the supplied file. These analytic levels are appropriate for an initial data-understanding workflow. Predictive analytics would require a model trained to predict a defined future or withheld outcome, whereas prescriptive analytics would require a justified mapping from evidence to action. Neither is claimed in the present experiment.

The distinction matters for interpreting the dashboard's support-oriented language. A screen that shows a low score or a threshold-based flag does not by itself estimate the probability of future failure. Predictive claims would require an outcome definition, temporally valid predictors, held-out evaluation, calibration and error analysis, and validation in another cohort. Similarly, a support recommendation is not proven helpful merely because it is plausible. The paper consequently uses terms such as "descriptive pattern," "observed association," and "candidate for review" rather than "cause," "risk prediction," or "impact."

C. Research Questions

The objectives are operationalized through five research questions. They keep the analysis aligned with the evidence available in the supplied dataset and dashboard artifacts.

D. Expected Contribution and Non-Claims

The intended contribution is practical rather than algorithmic novelty. The manuscript provides a reproducible descriptive analysis, connects each major dashboard view to a class of computations, consolidates closely related papers in one literature table, and records data-quality limitations that affect interpretation. It does not claim that Pearson correlation, linear regression, dashboard cards, heatmaps, or course summaries are newly invented methods. Nor does it establish that LearnTrack outperforms a baseline dashboard, reduces faculty workload, or improves student achievement. Those claims would require separate comparative and user-centered studies.

IV. Problem Statement and Objectives

Raw academic tables can make it difficult to inspect the distributions of attendance, daily study time, and examination outcomes together. LearnTrack addresses this by organizing key attributes into a set of descriptive and visual views. Its objectives are to: (1) summarize the provided data and check basic quality; (2) examine attendance, study-hour, and score distributions; (3) estimate linear associations among the primary variables; (4) compare summaries across course labels and performance categories; and (5) present these summaries in an interactive dashboard while making uncertainty and non-causal interpretation explicit.

V. System Overview and Methodology

A. Dashboard Implementation

The deployed prototype is available at https://https-github-com-deepika-deepu97-st.vercel.app/. Project materials identify a React/Vite frontend and Vercel deployment. The submitted screenshots show views labelled Dashboard, Students, EDA & Analytics, Correlations, Insights, and Reports. The dashboard shows cohort summary cards, scatter plots, descriptive statistics, performance distribution, course summaries, a correlation matrix, student listings, and report/export controls. The user interface contains an AI-synthesis section, but a screenshot alone does not establish a successful live AI service or validate the generated text; the paper therefore treats AI synthesis as an optional extension, not as an evaluated method.

B. Stakeholders and Intended Use

The primary intended user described in the project materials is faculty or an academic reviewer who needs a compact overview of a cohort. A faculty user can begin with the Dashboard page to understand the cohort at a glance, move to EDA and Analytics to examine distributions, use Correlations to inspect numeric relationships, compare course aggregates, and consult a student-record view when institutional permissions allow it. The Reports area is intended to make summary results shareable. These workflows are design intents inferred from the supplied screen labels and project-review slides; formal observation of users has not yet been performed.

The interface should not collapse distinct user needs into a single unrestricted view. Aggregate cohort analysis can often be performed without names or direct identifiers, whereas an authorized adviser may need identifiable rows to contact a student. For that reason, the proposed operational deployment should separate aggregate analytics from identifiable student lookup using role-based access. Export and report controls should apply the same permissions as on-screen student lists. A static public demonstration using illustrative records should be clearly distinguished from a production deployment connected to institutional records.

C. Functional Requirements and Acceptance Criteria

Table III documents requirements derived from the existing dashboard views and defines the evidence needed to verify each requirement. These are a test plan, not claims that all tests have already been completed. A visible chart is evidence that a UI component exists; numerical correctness requires comparison against a reproducible reference calculation, and usability requires evaluation with representative users.

TABLE II
RESEARCH QUESTIONS, EVIDENCE SOURCES, AND BOUNDARIES

ID	Research question	Evidence used	Boundary of the answer
RQ1	What are the distributions and basic quality characteristics of the supplied student dataset?	Schema inspection, counts, descriptive statistics, histograms, duplicate and missingness checks.	Answers describe this CSV only; provenance and representativeness are unknown.
RQ2	How are daily study hours and attendance associated with examination score in this file?	Pearson correlation, confidence intervals, scatter plots, and separate simple linear regressions.	Association is not causation; omitted variables and measurement quality are not controlled.
RQ3	How do summary outcomes vary across supplied course labels and study/attendance bands?	Group counts, group means and standard deviations, and banded descriptive tables.	Group differences are exploratory; course composition and assessment comparability are unknown.
RQ4	How clearly does the dashboard map its views to reproducible analysis?	Supplied screenshots, visible modules, and independently recomputed CSV results.	Interface presence and numerical traceability can be inspected; usability and educational impact are not measured.
RQ5	What must be resolved before the dashboard can support consequential institutional decisions?	Data quality audit, reference literature, privacy review, and identified validation gaps.	Produces a development and evaluation agenda, not a completed governance audit.

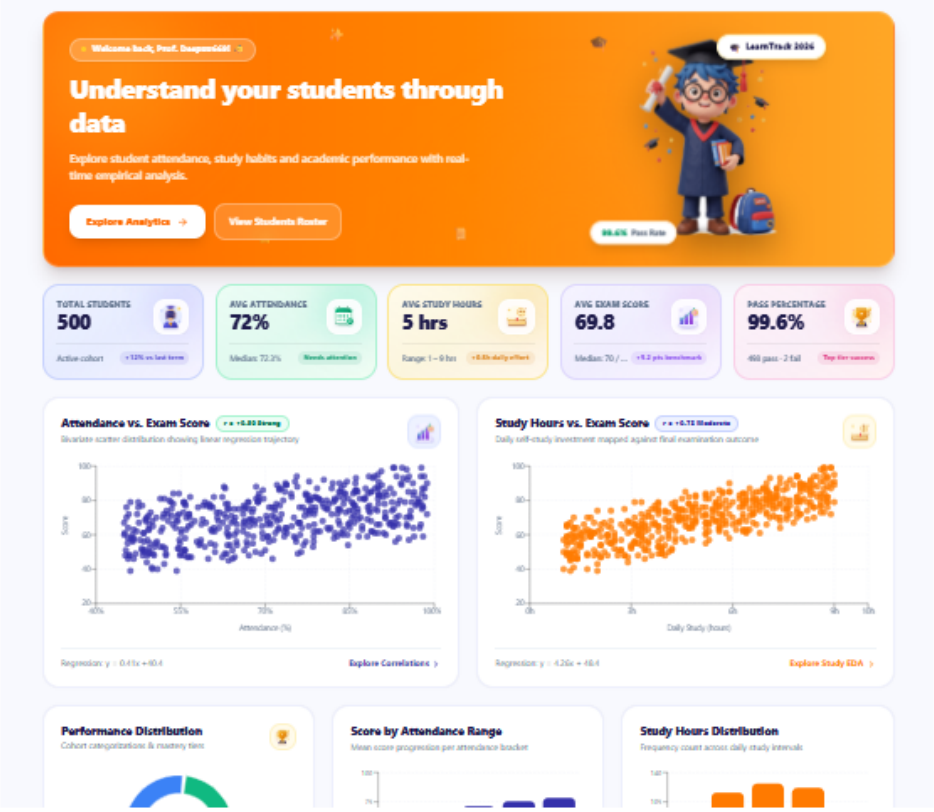


Fig. 1. LearnTrack dashboard screenshot supplied by the project team. The view shows cohort summary cards and attendance-versus-score and study-hours-versus-score plots. Dashboard-reported values are checked against the CSV separately in this paper.

D. Logical Architecture

At a conceptual level, the system can be described using four layers: data input, analytical computation, presentation, and governance. The data-input layer contains the CSV or an approved institutional feed; its contract includes field names, types, ranges, and provenance. The analytics layer computes descriptive summaries, bivariate associations, and course-level aggregates. The presentation layer renders values and plots in the React/Vite application. The governance layer surrounds every layer with versioning, data minimization, permissions, documentation, and review. This layering is a design model for the current prototype; it is not a claim that every control is implemented in the live service.

The project-review materials identify React/Vite, JavaScript, Python-based analysis, a charting layer, Firebase or relevant project storage, Vercel deployment, and Git/GitHub as technologies considered for the project. The present analysis verifies the CSV calculations and the supplied screenshots, but it does not inspect deployed source code, Firebase rules, authentication settings, or build configuration. Accordingly, this paper states the confirmed presentation/deployment information conservatively and treats the other named technologies as

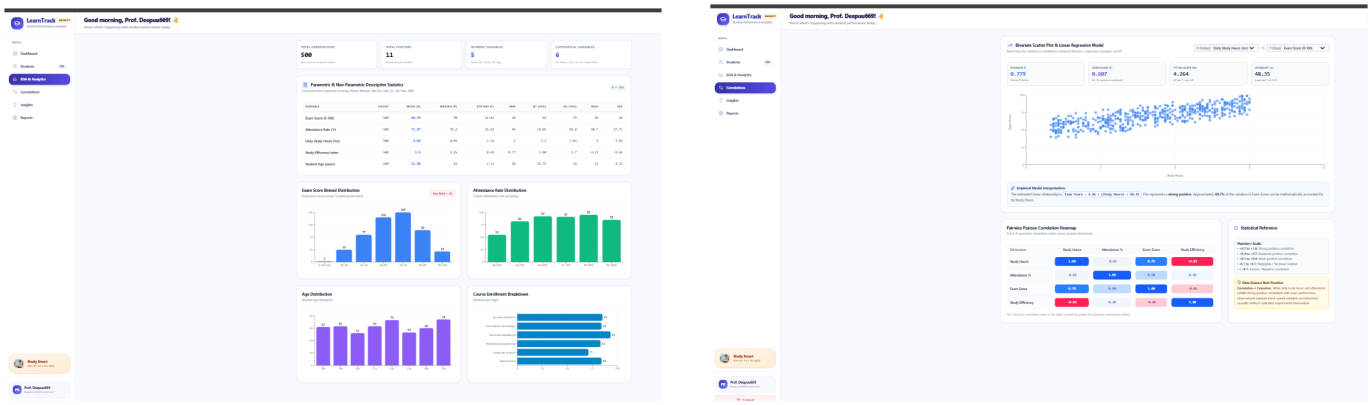


Fig. 2. Project screenshots supplied by the team: EDA and descriptive-statistics view (left) and correlation/regression view (right). These are interface evidence; the quantitative tables in this paper are calculated independently from the CSV.

TABLE III
FUNCTIONAL REQUIREMENTS AND VERIFICATION EVIDENCE

ID	Requirement	Acceptance evidence	Current status in this paper
FR-01	Show cohort-level summary values.	Recompute each card from a versioned CSV and compare within a documented rounding tolerance.	Partially checked; attendance display requires alignment to CSV result.
FR-02	Display distributions of study hours, attendance, and exam scores.	Compare plotted bins, counts, and axis labels with independently generated figures.	Offline figures are recomputed; UI screenshots are not pixel-by-pixel tested.
FR-03	Display correlations and fitted lines.	Recompute pairwise coefficients, regression slopes, intercepts, and R^2 from the same data version.	Statistical results are independently computed in the manuscript.
FR-04	Support course-level comparison.	Verify category labels and group sizes; display group means and within-group variability.	CSV group summaries are reported; course equivalence is not assumed.
FR-05	Provide student-record inspection.	Confirm access roles, minimum necessary fields, privacy protection, and behavior under unauthorized access.	UI view is visible in supplied artifacts; access-control behavior is not verified.
FR-06	Export or print summary reports.	Compare exported values with on-screen values and check whether direct identifiers are excluded by default.	Feature is present in the interface material; export contents require implementation testing.
FR-07	Offer generated analytical text, if enabled.	Record model/version, prompt, source metrics, hallucination checks, failure behavior, and user review pathway.	Optional feature only; not part of the evaluated quantitative method.

project-reported implementation choices or planned components until their active use is verified from the repository. This distinction prevents a project-plan item from being described as an implemented subsystem without code-level evidence.

E. Analytical Traceability as a Design Principle

Traceability means that each displayed quantitative result can be associated with a known source field, a formula, a filter or grouping rule, and a particular dataset version. It becomes particularly important if the system supports several views of the same data. For example, a course average, the overall mean, and a pass-rate card must all use consistent row inclusion and threshold rules. If a filtered view excludes observations, the filter should be visible or clearly indicated. If a dashboard uses a stored pre-computed metric, its refresh time and source version should be available to the operator.

A practical minimum is to store a data-version identifier, computation timestamp, metric name, source field, and calculation rule alongside exported reports. A more complete release process would include a small fixture dataset with known values, automated tests for metric functions, and a comparison report against the dashboard. The supplied screenshots reveal minor differences for attendance mean and attendance-score correlation. These are not large numerical differences, but they demonstrate why automated parity tests are preferable to visual inspection alone.

F. Analytical Workflow

The workflow is: load the CSV; inspect schema, missingness, duplicate rows, and identifier uniqueness; compute descriptive statistics; calculate Pearson correlations and simple least-squares regressions for the main variable pairs; produce aggregate charts; and display the results in the dashboard. No predictive classifier is trained in this paper.

G. Dataset and Fields

The provided file contains 500 records and 11 columns: `student_id`, `name`, `age`, `city`, `course`, `daily_study_hours`, `attendance_percent`, `exam_score`, `enrollment_date`, `performance_level`, and `study_efficiency`. It contains no missing values, no fully duplicated rows, and 500 unique student identifiers. Five fields are numeric and six are categorical/date-like fields, consistent with the dashboard's schema summary. The original source, sampling frame, collection process, and definitions of the supplied performance labels and study-efficiency feature were not provided. Although the file contains fields that resemble student records, its real-world versus synthetic origin

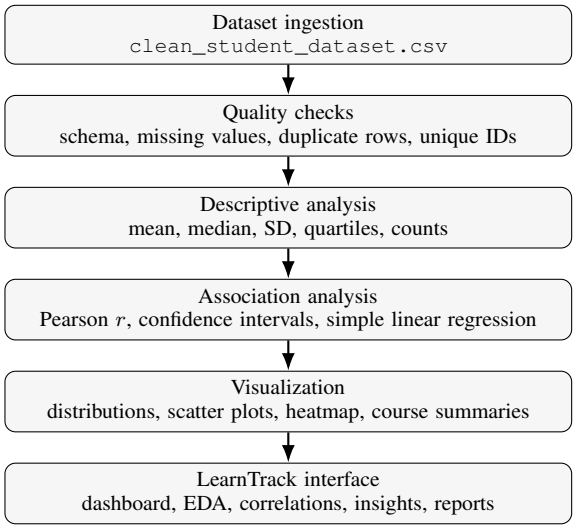


Fig. 3. Analysis and presentation workflow. The diagram represents the workflow reported in this paper, not a claim of a production data pipeline operating automatically.

TABLE IV
DATASET FIELD GROUPS

Fields	Role in analysis
`daily_study_hours`	Primary numeric variable; hours per day.
`attendance_percent`	Primary numeric variable; percentage scale.
`exam_score`	Primary outcome descriptor; score on a 0–100 scale.
`age`	Numeric demographic descriptor; summarized only.
`study_efficiency`	Numeric field supplied in CSV; definition unknown, excluded from primary claims.
`course`, `city`, `performance_level`	Categorical descriptors used for aggregate counts; categories are not independently validated.
`student_id`, `name`	Identifiers; not used as analytical predictors and omitted from reported individual-level results.
`enrollment_date`	Date-like field; not used in the primary analysis.

cannot be established from the file alone. Names and IDs are not reproduced in this paper.

H. Statistical Measures

For numerical variables, the sample mean, standard deviation, median, quartiles, minimum, and maximum were computed. Pearson correlation was used to summarize linear association for three pre-specified pairs. For paired observations (x_i, y_i), Pearson's coefficient is

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}. \tag{1}$$

Two-sided correlation p-values and approximate 95% confidence intervals via Fisher's z transformation were computed. Simple ordinary least-squares regression was used to summarize the line between each predictor and exam score; no causal interpretation is intended. Statistical tests are exploratory and are not adjusted for multiple comparisons. The score threshold of 40 is used only to reproduce the dashboard's pass-rate convention; any at-risk flag should be interpreted only after its rule is documented and validated.

I. Reproducibility and Data-Quality Protocol

The analysis script uses a fixed sequence so another researcher can reproduce the tables and plots: (i) load the comma-separated file with explicit column names retained from the header; (ii) check row and column counts, missing values, exact duplicate rows, and identifier uniqueness; (iii) parse the date-like enrollment field only for schema inspection; (iv) summarize numeric columns with count, mean, standard deviation, quartiles, and range; (v) compute pairwise Pearson correlations for specified variables; (vi) estimate separate simple linear regressions; and (vii) create the figures from the same in-memory DataFrame. The script does not use student names or identifiers as predictors, does not train a model, and does not replace missing observations because none were found in this file. This protocol is intentionally basic and auditable. The accompanying project archive contains the LaTeX source, the plotting script, the dashboard screenshots supplied by the project team, and the figures generated from the CSV. Tables with numerical values are transcribed from the same computed summaries used by the plotting script.

The terms "attendance," "daily study hours," and "exam score" are operationalized exactly as the column names in the file suggest: `attendance_percent`, `daily_study_hours`, and `exam_score`. These meanings are not independently verified against an institutional data dictionary. The `study_efficiency` feature is excluded from primary regression interpretation because its definition and units are undocumented. The `performance_level` category is counted as supplied but not treated as ground truth. This is important because category labels may be derived from examination scores; validating their independent relationship to scores would otherwise risk circular reasoning.

J. Operational Definitions and Data Audit

The variables are interpreted according to their supplied column names, not according to an external institutional data dictionary. This distinction affects measurement validity. For example, the variable `daily_study_hours` does not indicate whether hours were self-reported, automatically logged, averaged over weekdays, or recorded for a particular course. `attendance_percent` does not specify whether attendance is computed from all scheduled sessions or a selected period. `exam_score` is bounded between 0 and 100 in this file, but the assessment type, weighting, grading rubric, and comparability across courses are not documented. The analyses therefore use the values as present and do not impute unverified semantic detail.

K. Descriptive and Inferential Procedures

The analysis was conducted at the record level and summarizes all 500 rows in the supplied CSV. For a numeric variable

TABLE V
DATA AUDIT SUMMARY FOR THE SUPPLIED CSV

Audit item	Observed result	Interpretation and remaining check
Rows and columns	500 rows; 11 columns.	Confirms the size of the analyzed file, not the size of an underlying population.
Missing cells	0 missing values in the inspected file.	No imputation was required for the reported variables. Missingness outside this file or in upstream systems is unknown.
Exact duplicated rows	0.	Does not rule out duplicate people represented with differing identifiers or spelling.
Identifier uniqueness	500 unique values in `student_id`.	Unique IDs within this file do not verify person-level uniqueness across external systems.
Numeric bounds	Age 18–25; daily study hours 1–9; attendance 45.0–98.7; exam score 39–99.	No values exceed these observed ranges. The ranges are plausible for the file but are not validated business rules.
Categorical coverage	6 course labels; 4 performance labels; 15 city labels.	Category spelling and label semantics were accepted as supplied; category definitions should be documented.
Provenance and permissions	Not available in supplied material.	Must be documented before public release or use for consequential decisions.
Feature meaning	Formula for `study_efficiency` not provided.	Shown in numeric audit but excluded from substantive interpretation and recommendation logic.

X, the sample mean and sample standard deviation are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{2}$$

The median and quartiles are included because means and standard deviations alone do not show the full distribution. Minimum and maximum values are used as basic range checks, not as proof that the scale is valid. Counts and proportions describe the categorical labels, and each category denominator is the number of records assigned to that label in the supplied file.

Pearson's coefficient is reported for the three primary variable pairs. Its use gives a compact measure of linear association, but it can be influenced by outliers and does not capture every nonlinear pattern. Approximate confidence intervals for r use Fisher's transformation, $z = \operatorname{atanh}(r)$, with standard error $1/\sqrt{n-3}$ before transforming the interval back to the correlation scale. Two-sided p-values are reported descriptively; because several exploratory comparisons are made, they are not treated as confirmatory evidence. No multiplicity correction was applied. The results should therefore be read as a transparent description of the current file rather than as a pre-registered confirmatory experiment.

For each primary pair, ordinary least squares fits a line of the form $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. The slope gives the average change in the fitted response per one-unit change in the predictor within this dataset. The coefficient of determination R^2 expresses the in-sample fraction of response variance represented by the fitted one-predictor line. It is not a prediction accuracy score and is not evidence of out-of-sample performance. Residual assumptions, measurement error, confounding, sample selection, and independence of records have not been independently established. The linear fit is included to describe the visual trend and make the analysis reproducible, not to recommend a causal dose of study time or attendance.

L. Banded Summaries and Their Interpretation

To support inspection of group-level patterns, the supplementary analysis groups daily study hours into four bands (at most 3 hours, above 3 through 5 hours, above 5 through 7 hours, and above 7 hours) and attendance into four bands (at most 60%, above 60% through 75%, above 75% through 90%, and above 90%). These intervals are descriptive choices for readability. They are not official academic policy thresholds, learned cut-points, or validated risk categories. Records exactly at boundaries are assigned consistently by the code that produces each table and figure.

Binning is helpful when a stakeholder wants to compare mean scores across broad groups, but it loses information. Two students on opposite sides of a cut-point can have nearly identical measured values while appearing in different bands; two students in the same band may still have very different attendance or study times. Banded means should therefore be read alongside continuous scatter plots and distribution plots. The analysis does not use band membership to train a model or to label a person as at risk.

M. Analysis Environment and Reproducibility Scope

The associated project package contains LaTeX source, plotting scripts, figures, and dashboard screenshots. The CSV itself is supplied separately so that it is not inadvertently included in a public source archive. The numerical outputs in this manuscript are intended to be reproduced from the same file version. For a publication release, the authors should archive a checksum of the data file, Python and package versions, the analysis command, and generated output logs. The current package documents the expected input schema and the figures used in the manuscript, but the deployed web application has not been subjected to a formal software verification campaign.

A minimal reproducibility record should include: (1) the exact CSV filename and checksum; (2) the number of rows and columns after loading; (3) the column names and data types; (4) the missing-value, duplicate-row, and unique-ID counts; (5) summary statistics and group sizes; (6) the statistical functions and confidence-interval convention; (7) the versions of the libraries used; and (8) the commit or release identifier for the dashboard. These items permit reviewers to distinguish a real change in the data from a rounding change, a stale screenshot, or a software regression.

TABLE VI
DESCRIPTIVE STATISTICS (N=500)

Variable	Mean	SD	Median	Min	Max
Age (years)	21.58	2.33	22.00	18	25
Study hours/day	5.03	2.34	4.95	1.00	9.00
Attendance (%)	71.91	15.79	72.10	45.00	98.70
Exam score	69.79	12.83	70.00	39	99
Study efficiency*	1.50	0.69	1.25	0.77	4.13

*Feature definition and unit not supplied; shown for completeness only.

TABLE VII
SUPPLIED PERFORMANCE LABELS AND PASS-RATE SUMMARY

Category / measure	Count	Share
Poor	37	7.4%
Average	136	27.2%
Good	207	41.4%
Excellent	120	24.0%
Score ≥ 40 (pass convention)	498	99.6%
Score < 40	2	0.4%

TABLE VIII
QUANTILES OF THE PRIMARY VARIABLES

Variable	P10	Q1	Median	Q3	P90	P95
Study hours/day	1.70	3.20	4.95	7.03	8.30	8.70
Attendance (%)	49.40	58.05	72.10	85.80	93.10	96.01
Exam score	51.00	61.00	70.00	79.00	87.00	90.00

VI. RESULTS

A. Descriptive Statistics

Table VI summarizes the main numerical variables. Mean examination score was 69.79, while the median was 70.00. Mean daily study time was 5.03 hours and mean attendance was 71.91%. Two records had scores below 40; under a threshold of 40, the pass rate was therefore 99.6%. The supplied dataset's performance labels count 37 records as Poor, 136 as Average, 207 as Good, and 120 as Excellent. Because the label-generation rule is undocumented, these labels are displayed as provided rather than treated as independent grading validation.

B. Distributional Detail and Quantile Interpretation

The mean score of 69.79 is close to the median of 70, which indicates that the center of the score distribution is not strongly shifted by a large difference between these two location measures. The central half of scores spans the first and third quartiles, 61 and 79, respectively. Study hours have a median of 4.95 hours per day; the middle 50% of recorded values span 3.20 to 7.025 hours. Attendance has a median of 72.10%, with the interquartile range from 58.05% to 85.80%. These summaries provide a fuller view than the mean alone and should be revisited if additional data or a documented sample frame becomes available.

A quantile is a rank-based summary rather than a guarantee of how a future cohort will behave. In this file, the 10th percentile score is 51 and the 90th percentile is 87, so the observed middle 80% of scores spans 36 points. That spread reinforces why an overall average should not be used to infer the status of every individual. Likewise, attendance ranges from 45.0% to 98.7% and study hours range from 1 to 9 hours per day. A dashboard should display these distributions and avoid replacing them with a single label wherever the wider spread is relevant.

C. Correlation and Regression Results

Table IX reports the primary associations. Daily study hours correlate positively with exam score ($r = 0.779$), and the fitted simple regression has slope 4.264 points per additional reported study hour with intercept 48.349. Attendance has a moderate positive correlation with score ($r = 0.502$) and a fitted slope of 0.408 exam-score points per attendance percentage point. Study hours and attendance are weakly and non-significantly associated ($r = -0.055$, $p = 0.219$). Slope values describe the fitted lines within this dataset and should not be interpreted as expected causal effects.

D. Course-Level Summaries

Table X shows the mean score, mean attendance, and mean daily study time for the six course labels. Figure 9 plots course means with 95% confidence intervals; these intervals describe sampling uncertainty under common assumptions and do not correct for possible differences in course composition or assessment difficulty. Computer Science has the highest mean score (70.76) in this sample, while Data Science has the lowest (68.38). The score differences are small compared with the spread of individual scores and should not be interpreted as stable course rankings: the dataset source, sample selection, curriculum, and assessment equivalence are unknown.

E. Correlation Matrix Across Numeric Fields

Table XI extends the primary analysis to every numeric field in the supplied file. The strong negative association between study hours and the study-efficiency column ($r = -0.835$) is not interpreted substantively because the definition of study efficiency is not available. The matrix helps disclose this data structure rather than silently excluding a result that might affect downstream use.

F. Stratified Descriptive Analysis

To improve interpretability, the continuous measures were also summarized within simple bins. Binning is used only for descriptive communication; it loses information and does not constitute a predictive model. Table XII shows a monotonic pattern in the mean score across the selected study-hour brackets in this sample. Table XIII shows a similar gradient across attendance brackets. As these are unadjusted observational summaries, neither table demonstrates that increasing time or attendance alone would produce the displayed difference. Differences may reflect other variables, dataset construction, or how the score was generated.

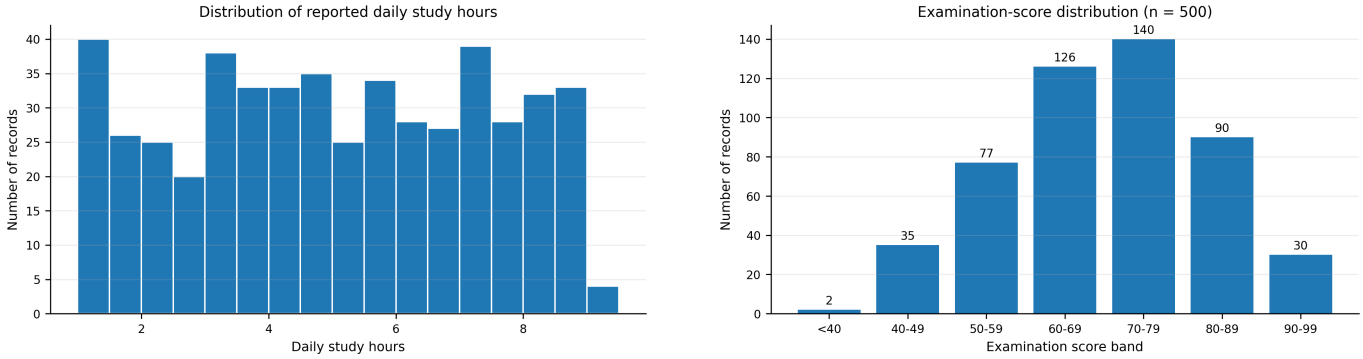


Fig. 4. Univariate distributions computed from the supplied CSV: daily study hours (left) and examination-score bands (right).

TABLE IX
PAIRWISE PEARSON CORRELATIONS AND SIMPLE LINEAR REGRESSION SUMMARIES

Pair	**Pearson r**	**95% CI for r**	**p-value**	**OLS slope/intercept**	R^2
Study hours vs. exam score	0.779	[0.742, 0.811]	< 0.001	$y = 4.264x + 48.349$	0.607
Attendance vs. exam score	0.502	[0.433, 0.565]	< 0.001	$y = 0.408x + 40.469$	0.252
Study hours vs. attendance	-0.055	$[-0.142, 0.033]$	0.219	$y = -0.371x + 73.772$	0.003

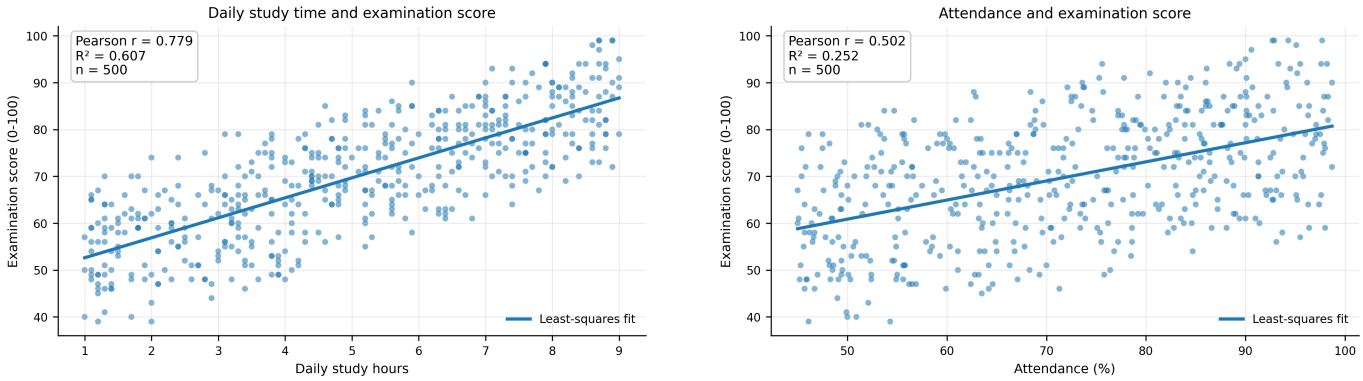


Fig. 5. Scatter plots and fitted simple linear regression lines for the two primary factors and examination score.

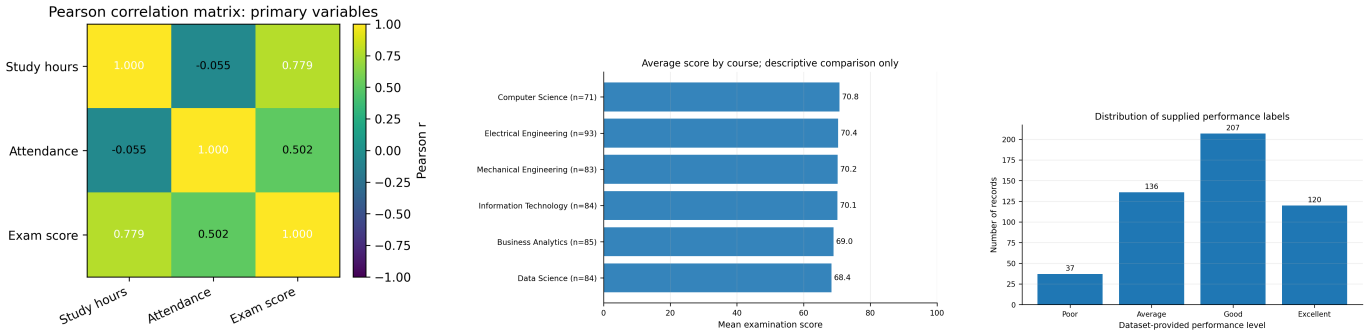


Fig. 6. Correlation matrix for the three primary variables (left), mean examination score by course (middle), and category counts for the supplied performance labels (right).

G. Study-Time and Attendance Band Comparisons

Banded group means provide another way to inspect the observed relationships without treating the bands as predictions. The mean examination score increases across the study-time bands in the current CSV: 56.33 for the at-most-three-hour group, 65.50 for the next band, 74.14 for the third band, and 82.61 for records reporting more than seven hours per day. The number of records in each band ranges from 116

TABLE X
COURSE-LEVEL DESCRIPTIVE SUMMARY

Course	n	Mean exam score	Mean attendance (%)	Mean study hours/day	Pass rate (score ≥ 40)
Business Analytics	85	69.00	70.72	5.07	98.82%
Computer Science	71	70.76	73.62	4.94	100.00%
Data Science	84	68.38	71.96	4.72	100.00%
Electrical Engineering	93	70.37	70.45	5.23	100.00%
Information Technology	84	70.14	71.95	4.97	98.81%
Mechanical Engineering	83	70.20	73.21	5.21	100.00%

TABLE XI
PEARSON CORRELATION MATRIX FOR ALL NUMERIC FIELDS

	Age	Study hours	Attendance	Exam score	Study efficiency
Age	1.000	-0.062	-0.025	-0.076	0.031
Study hours	-0.062	1.000	-0.055	0.779	-0.835
Attendance	-0.025	-0.055	1.000	0.502	0.302
Exam score	-0.076	0.779	0.502	1.000	-0.464
Study efficiency	0.031	-0.835	0.302	-0.464	1.000

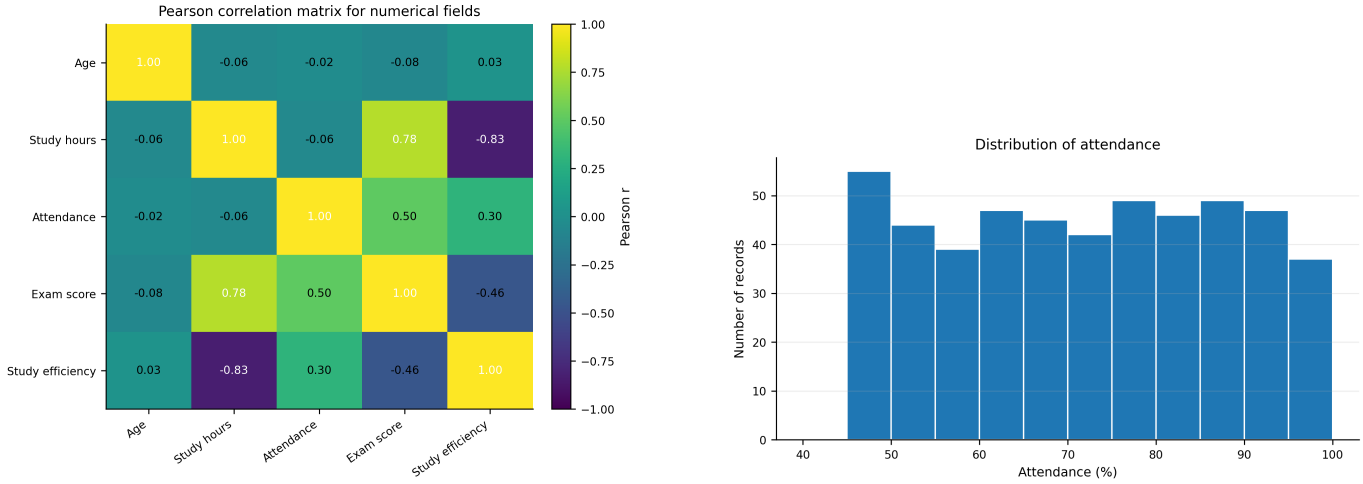


Fig. 7. Figures generated from the supplied CSV: correlation matrix across the five numeric fields (left) and attendance distribution (right). The study-efficiency column is shown for transparency, although its definition is unknown.

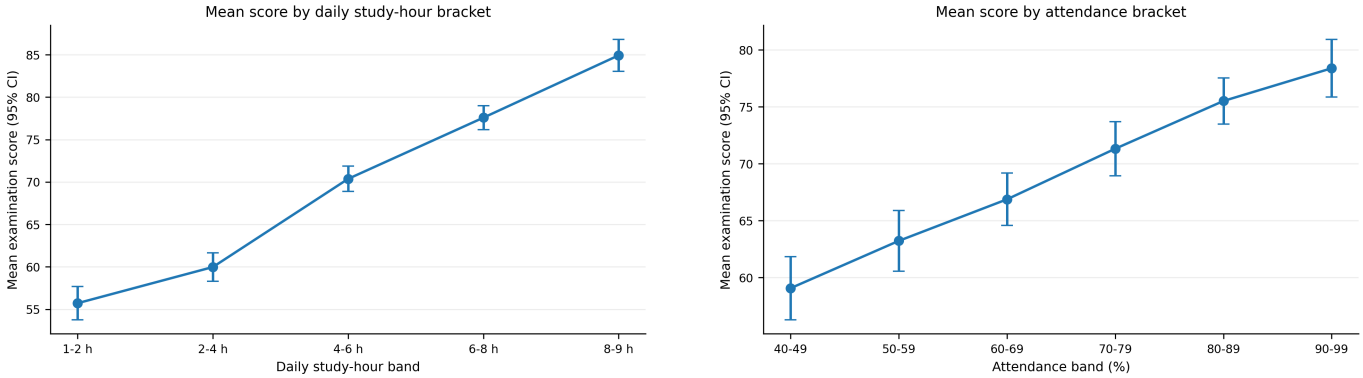


Fig. 8. Group means with 95% confidence intervals by daily study-hour band (left) and attendance band (right). Bins are descriptive and do not adjust for confounders.

to 135. This ordered pattern is consistent with the positive Pearson association, but it cannot determine whether more study time itself caused the score differences. Prior performance, course difficulty, self-selection, reporting accuracy, and the time window over which study hours were measured may contribute.

TABLE XII
MEAN EXAM SCORE BY DAILY STUDY-HOUR BRACKET

Hours/day	n	Mean score	95% CI
1–2	66	55.71	[53.76, 57.66]
2–4	116	59.97	[58.31, 61.64]
4–6	127	70.37	[68.88, 71.86]
6–8	122	77.58	[76.15, 79.01]
8–9	69	84.93	[83.04, 86.81]

TABLE XIII
MEAN EXAM SCORE BY ATTENDANCE BRACKET

Attendance	n	Mean score	95% CI
40–49%	55	59.06	[56.28, 61.83]
50–59%	83	63.22	[60.55, 65.88]
60–69%	92	66.87	[64.56, 69.18]
70–79%	91	71.32	[68.93, 73.71]
80–89%	95	75.52	[73.48, 77.55]
90–99%	84	78.39	[75.85, 80.93]

Mean scores also increase across the attendance bands, from 61.68 in the group with attendance at most 60% to 78.25 in the group above 90%. Each band contains a different number of records, and the highest-attendance group includes 80 records. A banded result is an aggregate summary: it does not mean every record in a higher band achieved a higher score than every record in a lower band. The within-band standard deviations remain substantial, so the group mean should not be converted into an individual prediction.

The increase in band means is useful for exploratory communication, but it can also create an overly simple story if the underlying variability is hidden. The table does not adjust for other variables, compare standardized assessment difficulty, or estimate an intervention effect. For a later predictive task, band selection would need to be justified on a training dataset and evaluated on untouched data. For the current purpose, continuous measures remain the preferred inputs to the main association analysis, and the bands are only a visual summary aid.

H. Course Comparisons and Supplied Categories

The six course labels have mean examination scores between 68.38 and 70.76, a range of approximately 2.38 points. The standard deviations within the courses are roughly 11.78–13.78 points, larger than the difference in course means. An exploratory one-way ANOVA comparing course means yields $F(5, 494) = 0.414$, $p = 0.839$; the assumptions and representativeness of the dataset are not established, so this result is reported only as a descriptive check and not as evidence of academic equivalence. Figure 9 presents means with approximate 95% intervals.

I. Dashboard Features and Analytical Traceability

Table XVI maps each visible dashboard area to the computations described in this study. This traceability is important because the web interface contains more than one kind of displayed result: descriptive summaries can be directly recomputed from a CSV, whereas user-interface recommendations or AI-generated summaries need additional validation. The table separates current analytical support from future evaluation needs.

J. Consistency Check Against Dashboard Screenshots

The dashboard screenshots provided with the project show 500 students, a mean examination score of 69.8, average daily study time of about 5 hours, a pass percentage of 99.6%, and positive study-hour and attendance relationships with score. The score mean, study-time mean, pass convention, and study-hour correlation agree with independent calculations from the CSV after rounding. Two display-level differences require reconciliation: the supplied dashboard screenshot reports average attendance as 71.97% whereas the CSV mean is 71.9074% (71.91% rounded); and the screenshot shows attendance–score correlation around 0.5044 whereas recomputation gives 0.50196 (0.502 rounded). In addition, the screenshot displays 137 students flagged for support, while the exact flagging rule is not documented in the submitted evidence. The manuscript reports reproducible CSV-derived calculations and does not treat the dashboard's support count as a validated classifier result.

K. Uncertainty, Interpretation, and Practical Use

Pearson's coefficient measures linear covariation, not whether a student's additional study hour will cause an increase in examination score. In this dataset, the fitted line for study time is $\widehat{score} = 48.349 + 4.264\,hours$, while the attendance line is $\widehat{score} = 40.469 + 0.408\,attendance$. These simple one-predictor regressions share unmodeled variation with all other observed and unobserved characteristics. The R^2 values indicate how much in-sample variation is represented by each fitted line, but do not indicate out-of-sample prediction accuracy. No train/test split is necessary for the descriptive questions addressed here, and no classification metric is appropriate because no predictive classifier was trained.

For faculty, the intended use is exploratory navigation: identify aggregate patterns, review distributional spread, compare course summaries, and then investigate records using institutional procedures. The dashboard should not be used as a stand-alone decision system for penalties, grading, admission, or progression. If a student-support flag is operationally necessary, the institution should publish the rule, validate it against an outcome measured independently from the threshold itself, evaluate false alarms and missed cases, and provide a human review path. These requirements are consistent with the literature's emphasis on the interpretation and responsible use of analytics rather than the mere generation of a score or alert [31]–[33], [43].

VII. DISCUSSION

The results illustrate how several types of information can be made available in a single interface: central tendencies and spread, distributions, associations, course-wise summaries, attendance- and study-hour-band summaries, and supplied performance labels. The banded summaries were added to make

TABLE XIV
MEAN SCORE BY STUDY-HOUR AND ATTENDANCE BANDS

(a) Daily study hours

Band (hours/day)	n	Mean score	SD
≤ 3	116	56.33	8.18
> 3 to 5	135	65.50	9.24
> 5 to 7	124	74.14	8.24
> 7	125	82.61	8.14

(b) Attendance

Band (%)	n	Mean score	SD
≤ 60	139	61.68	11.65
> 60 to 75	133	68.11	11.36
> 75 to 90	148	74.35	10.84
> 90	80	78.25	11.65

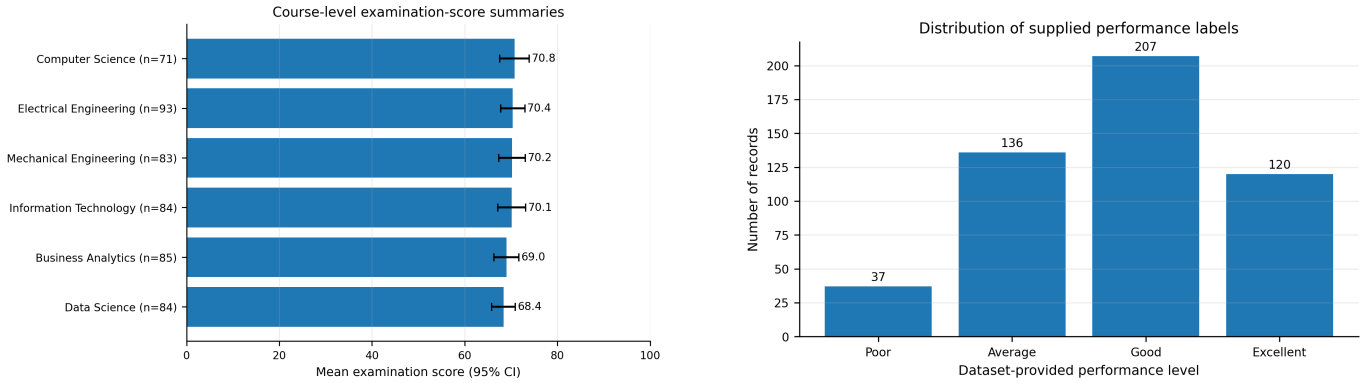


Fig. 9. Course-level mean examination scores with 95% confidence intervals (left) and counts for the dataset-provided performance labels (right). Label definitions were not supplied, so label counts are not treated as a validated assessment scale.

TABLE XV
COURSE-LEVEL RESULTS RECOMPUTED FROM THE SUPPLIED CSV

Course	n	Mean score	SD	Mean attendance	Mean study hours	Pass convention
Business Analytics	85	69.00	12.49	70.72%	5.07	98.82%
Computer Science	71	70.76	13.41	73.62%	4.94	100.00%
Data Science	84	68.38	11.78	71.96%	4.72	100.00%
Electrical Engineering	93	70.37	12.65	70.45%	5.23	100.00%
Information Technology	84	70.14	13.78	71.95%	4.97	98.81%
Mechanical Engineering	83	70.20	13.09	73.21%	5.21	100.00%

TABLE XVI
DASHBOARD VIEW-TO-ANALYSIS MAPPING AND VALIDATION STATUS

Dashboard view	Displayed analytical content	Validation status / interpretation
Dashboard	Cohort size, mean attendance, mean study hours, mean score, pass-rate card, scatter plots	Summary values can be independently recomputed. Some dashboard figures differ slightly from the CSV and should be aligned before publication.
Students	Student table with ID, course, attendance, study hours, score, and supplied category	Operational student-level screen. This research paper does not publish identifiable rows. Privacy and access control require separate review.
EDA and Analytics	Descriptive summary, score bins, attendance bins, age and course distributions	The present paper recomputes the summaries and provides the figures and tables in the Results section.
Correlations	Pearson coefficients, fitted line, R^2, correlation heatmap	The paper calculates coefficients from all 500 rows and reports confidence intervals and p-values.
Insights	Rule-based statistical findings and optional AI synthesis panel	Rule-based statements can be checked against computed values; successful AI operation and educational value are not established by screenshots.
Reports	Executive summary, descriptive metrics, correlations, course comparison, export/print controls	This paper provides corresponding summary tables. The consistency check documents differences from some screenshot values.

associations easier to interpret, but they should not replace the continuous-variable scatter plots because binning can hide variability and create artificial boundaries. The strongest measured association in the CSV is between daily study hours and examination score; attendance also has a positive association with score. The lack of a notable association between study hours and attendance in the same file suggests that the two variables provide different descriptive views of the cohort. These findings are consistent with studying attendance and study time as related but non-identical dimensions [14], while they cannot establish the direction of causation or rule out confounding.

The interface is aligned with the general learning-dashboard design goals discussed in the literature: show relevant educational data, make patterns interpretable, and provide a useful overview for a stakeholder [22]–[27]. However, a successful implementation is not evidence that instructors make better decisions, that the dashboard improves student outcomes, or

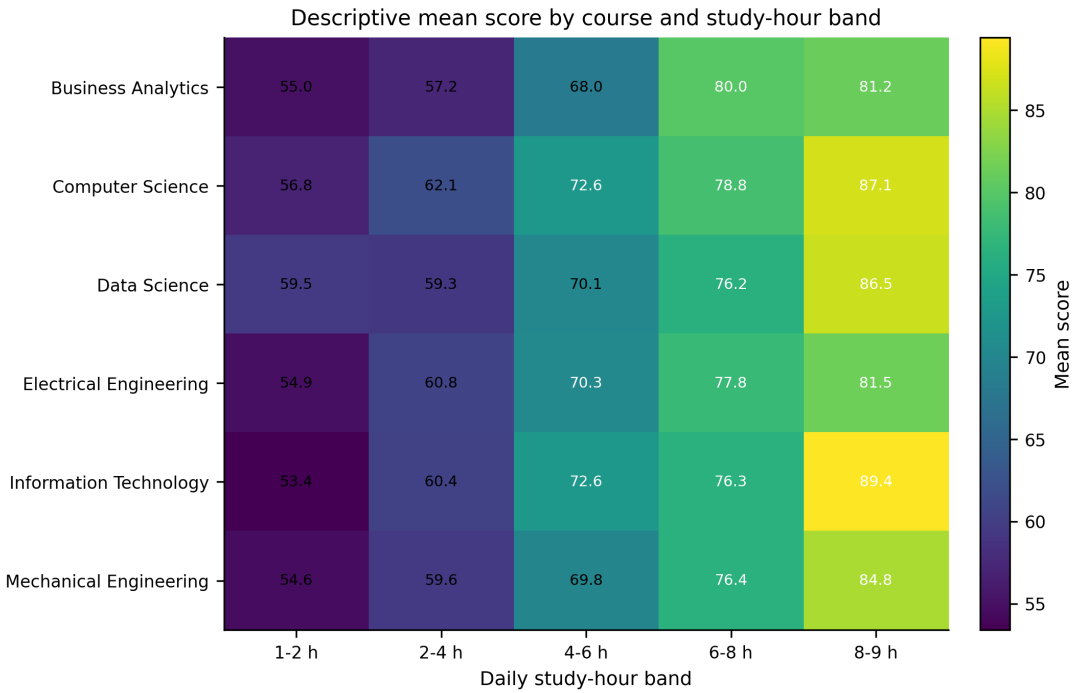


Fig. 10. A course-by-study-hour heatmap of mean examination scores. Empty cells, if any, represent a course–band combination without observations; no value should be inferred for an empty cell. This chart supports exploration of possible differences in study-hour patterns across course labels.

that it is usable for all groups. Such outcomes require separate user studies and field evaluation.

A. Answers to the Research Questions

RQ1 is answered by the data audit and distribution tables. The inspected file contains 500 rows and 11 fields, with no missing cells, no exact duplicate rows, and 500 unique student IDs. The mean score is 69.79 (SD 12.83), mean attendance is 71.91%, and mean daily study time is 5.03 hours (SD 2.34). These results provide a reproducible description of the file, but not evidence that the sample represents any particular institution, course population, or time period.

RQ2 is answered by the pairwise association analysis. Study hours and examination score have a strong positive linear correlation in the supplied file ($r = 0.779$), while attendance and score have a moderate positive correlation ($r = 0.502$). The study-hours/attendance coefficient is small and not statistically significant ($r = -0.055$, $p = 0.219$). The data therefore show different descriptive relationships for study hours and attendance, but do not establish a causal mechanism. The absence of a strong correlation between study time and attendance also does not imply that the variables are independent in the population; it is a statement about the present file and the chosen linear measure.

RQ3 is answered through course summaries and the two banded comparisons. Course mean scores range from 68.38 to 70.76, while within-course standard deviations are considerably larger. The exploratory course comparison should not be interpreted as proof that course outcomes are equivalent, because course composition, assessment difficulty, and sampling conditions are not documented. The banded tables reveal ordered group means for study time and attendance, but those aggregate differences are not individual-level predictions and are not used to define a validated at-risk rule.

RQ4 is answered partially. The screenshot evidence shows a multi-view interface containing cohort summaries, EDA views, correlation displays, student records, insights, and reports. Independently recomputed calculations supply a reference against which the interface can be checked. Two small mismatches are documented for attendance mean and attendance-score correlation, and the rule behind a support flag is not provided. Therefore, interface coverage is visible and some metric traceability is possible, but overall numerical parity, usability, reliability, accessibility, and user impact remain to be tested.

RQ5 yields a concrete set of prerequisites: document the dataset source and permissions; define the meaning and calculation of each feature; reconcile all displayed metrics against the analysis code; implement and test role-based access; minimize the use of direct identifiers; and evaluate any support or AI-generated recommendation before consequential use. These priorities are not optional extras for a mature educational analytics tool. They determine whether the tool's output can be reproduced, understood, and responsibly acted upon.

B. Implications for Dashboard Design

First, metric definitions should be visible and consistent. Tooltips or a short methodology panel can explain the denomina-

tor for attendance, the scale for exam score, the period for study time, and the threshold used for any pass-rate or support card. Second, visual views should complement one another: a mean-value card communicates the center, a histogram shows spread, a scatter plot shows the relationship between two continuous variables, and a group comparison summarizes a categorical split. No one graphic answers all of these questions. Third, a dashboard should make data limitations discoverable near the point where a user might overinterpret a result. A note in a research paper may not reach an instructor who only sees the deployed interface.

Fourth, recommendations should be phrased as prompts for investigation rather than labels about student ability. An aggregate difference can suggest where an adviser might ask further questions, but a label like "poor performer" may be stigmatizing and may obscure contextual reasons for an observed score. Fifth, every export should preserve the source date, metric definition, and any filtering rules. These practices improve interpretability and make it easier to reconcile outputs across the dashboard, a paper, and a later institutional report.

C. Implications for Research Practice

The project shows that a small, interpretable analysis can be useful as a first step before considering machine learning. Descriptive methods reveal data-quality issues, unexpected ranges, category counts, and variable relationships that need to be understood before attempting prediction. Building a classifier before clarifying the target definition, checking for leakage, or validating feature meaning could produce persuasive but untrustworthy performance metrics. For a student-support use case, an interpretable baseline and a transparent decision process should precede more complex models.

A next research iteration should pre-specify an evaluation protocol. For the software itself, tasks might include locating the mean score, identifying a group with comparatively lower aggregate attendance, and correctly interpreting the correlation heatmap. Measures could include task completion, completion time, interpretation accuracy, and the frequency of incorrect causal conclusions. For a later intervention study, a separate design would be needed to determine whether access to the dashboard changes advisor behavior or student outcomes. Such an impact study should include a suitable comparison condition, defined follow-up period, ethical approval as required, and reporting of both positive and null results. It should not infer impact from usage logs or user satisfaction alone.

VIII. Limitations, Ethics, and Future Work

First, the provenance of the dataset is unknown: no publisher, institution, collection method, sampling frame, or usage permission was provided. Before publication, the authors should document whether the file is synthetic, public, or institutionally collected and verify that its use is permitted. Second, a single snapshot cannot establish temporal or causal relations. Third, the definitions of `performance_level` and `study_efficiency` were not provided; the latter is excluded from the core inference, while the former is only reported as a supplied category. Fourth, dashboard values for attendance and the support flag differ from straightforward CSV re-computation and must be harmonized. Fifth, there was no formal user study, baseline comparison, software latency benchmark, or independent replication in the supplied evidence.

Student information requires privacy-aware handling. This paper reports only aggregates and excludes names and identifiers from figures and tables. Before release, the team should remove or synthesize identifying fields, obtain required permissions, consider re-identification risk in small course groups, and document whether informed consent or institutional review is needed. An at-risk view should be described as a triage aid only after its thresholds are explicit, transparent, and assessed for errors and group disparities.

Future work should document provenance, create a reproducible analysis script, reconcile the UI and offline metrics, and replicate results with a documented independent dataset. The supplied `study_efficiency` column requires a formula, scale, and justification before it can be interpreted or used analytically. The at-risk rule should be formalized as a transparent rule, and its false-positive/false-negative rates should be measured against an independent and ethically approved outcome definition. A dashboard deployment audit should also cover role-based access, export controls, data retention, and whether names or identifiers are necessary for each user workflow. A user study with faculty could measure task completion, interpretation accuracy, and perceived usefulness. If predictive analytics are later added, the model should be evaluated separately using leakage-aware train/test procedures, appropriate metrics, calibration, subgroup analysis, and a clear decision-support policy. Predictive results should not be inferred from the correlations reported here.

IX. System Verification and Evaluation Plan

The current paper presents a research prototype and an offline analysis, not a completed software validation. The following test matrix translates the identified risks into verifiable activities for a subsequent project release. The owner and completion date for each item should be set by the project team once the source repository and deployment configuration are available.

These tests should be prioritized by consequence. Numeric consistency and protection of identifiable records are foundational. Usability, accessibility, AI-output checks, and impact evaluation then establish whether the tool can be used safely and effectively for the intended tasks. Passing a functional test does not demonstrate that the dashboard improves learning; this distinction should remain in the project's documentation and promotional materials.

X. Conclusion

This paper presented LearnTrack, an exploratory student performance analytics dashboard and an empirical description of a supplied dataset containing 500 records and 11 fields. The CSV-derived mean exam score was 69.79, mean daily study time was 5.03 hours, and mean attendance was 71.91%. Study hours showed a strong positive linear association with

TABLE XVII
PROPOSED VERIFICATION AND EVALUATION PLAN

Test ID	Risk	Verification procedure	Evidence to retain
T-01	Stale or different data versions.	Freeze one CSV file, record its checksum, recompute all metrics, and confirm the dashboard consumes the same version.	Checksum, dataset version, analysis log, release tag.
T-02	Arithmetic or rounding defects.	Use hand-checkable fixture rows with known means, percentages, correlations, and boundary values; compare output to expected values.	Automated unit-test report and fixture definition.
T-03	Incorrect chart labels or grouping.	Test minimum, maximum, missing categories, boundary values, and empty course-band combinations.	Screenshot and machine-readable chart data.
T-04	Confusing correlation with causation.	Ask representative users to interpret scatter plots and correlations using standardized scenarios.	User responses, scoring rubric, common error summary.
T-05	Unauthorized viewing or export of identifiable records.	Test least-privilege roles and direct access to protected routes, datasets, and report downloads.	Access-control test report and policy configuration.
T-06	AI-generated statement unsupported by source metrics.	If the AI feature is enabled, compare each generated claim against a metric ledger and test timeout, empty context, and failure states.	Prompt/model version, inputs, outputs, error log, audit scores.
T-07	Accessibility or responsive-layout failures.	Test keyboard navigation, contrast, text scaling, mobile viewport widths, and screen-reader labels.	Accessibility checklist and representative screenshots.
T-08	Unsupported claims of educational impact.	Keep a separate impact-study protocol with target outcome, comparison, timing, inclusion criteria, and analysis plan.	Approved protocol and final preregistered measures where applicable.

exam score ($r = 0.779$), and attendance showed a moderate positive association ($r = 0.502$). Study hours and attendance were not significantly associated ($r = -0.055$, $p = 0.219$). The system brings summary statistics, charts, correlation analysis, and course summaries into one interface. These are descriptive results; the dataset provenance, dashboard metric mismatches, and absence of a formal user evaluation are important limitations. Resolving these issues and replicating the analysis on a documented dataset are necessary next steps before making broader educational or publication claims.

APPENDIX A
VARIABLE DICTIONARY AND ANALYTICAL TREATMENT

Table XVIII records the observed field names and how each is handled in this manuscript. The descriptions below are operational interpretations of names in the CSV. They should not be mistaken for definitions certified by a registrar or institution. The original dataset documentation should supersede these provisional descriptions if it becomes available.

The most important dictionary gaps are the study-time reporting period, attendance denominator, grade definition, performance-category rule, and study-efficiency formula. These are not merely editorial omissions. Without these definitions, a researcher cannot establish whether two data extracts are comparable or whether a derived variable duplicates information from the target. For instance, if `performance_level` is generated directly from `exam_score`, the association between the two would be circular rather than independent validation. The manuscript therefore does not claim that the supplied labels validate the numerical score.

APPENDIX B
REPRODUCIBLE ANALYSIS SEQUENCE

The analysis can be repeated by following a fixed sequence. This pseudocode documents the logic of the workflow without claiming that every line is the exact content of a deployed application's source code.

1) Load the CSV while preserving the column names from the file header.
2) Verify the file has 500 records and the expected 11 columns; check data types and parse numeric values.
3) Calculate the number of missing values, exact duplicate rows, and unique student identifiers.
4) Generate numeric summaries: count, mean, standard deviation, median, quartiles, minimum, and maximum.
5) Calculate the pairwise Pearson coefficients for study hours versus score, attendance versus score, and study hours versus attendance.
6) Estimate two-sided correlation tests and approximate 95% confidence intervals using Fisher's transformation.
7) Fit a separate one-predictor ordinary least-squares line for each primary pair and record its slope, intercept, and in-sample R^2.
8) Aggregate scores, attendance, and study time by course and by the pre-defined descriptive bands.
9) Generate plots and result tables from the same in-memory data frame; record the dataset version and analysis environment.
10) Compare the offline output against any dashboard card or chart that reports the same metric, then document discrepancies instead of manually editing the expected result.

A release-quality reproduction should add a requirements file or lockfile, library versions, a data checksum, an environment description, and an automatically generated results log. The input CSV should remain outside a public archive unless the authors can document that sharing is permitted. If the dataset is synthetic, that fact should be stated explicitly; if it contains real student records, an appropriate anonymization, legal basis, and review of re-identification risk are required before distribution.

APPENDIX C
INTERPRETATION CHECKLIST FOR READERS

The following checklist helps readers distinguish a supported result from an unsupported inference.

- A mean or median describes the observed sample; it does not define an individual student's typical future score.

TABLE XVIII
VARIABLE DICTIONARY FOR CLEAN_STUDENT_DATASET.CSV

Field	Observed type	Provisional role	Treatment in this paper
student_id	Text	Record identifier.	Checked for within-file uniqueness; not used as an analytical predictor and not listed in result tables.
name	Text	Potential direct identifier.	Excluded from statistical analysis and not reproduced in the manuscript.
age	Integer	Age-like numeric descriptor.	Summarized and included in the numeric correlation matrix only.
city	Category/text	Location label.	Counted as a field and retained for the schema audit; no geographic or demographic inference is made.
course	Category/text	Course/programme label.	Used for group counts and descriptive comparisons; equivalence of course assessments is not assumed.
daily_study_hours	Decimal numeric	Reported daily study-time quantity.	Primary predictor in a descriptive association with exam score; recording method and period are undocumented.
attendance_percent	Decimal numeric	Attendance percentage.	Primary predictor in a descriptive association with exam score; denominator and time window are undocumented.
exam_score	Integer numeric	Assessment score, apparently on a 0–100 scale.	Main outcome descriptor; grading method and assessment comparability are undocumented.
enrollment_date	Date-like text	Enrollment-date field.	Examined as part of the schema; not used for the primary statistical claims.
performance_level	Category/text	Supplied performance label.	Counts are reported as supplied. The label-generation rule is unknown, so it is not treated as independent ground truth.
study_efficiency	Decimal numeric	Numeric feature with undocumented meaning.	Described in the numeric audit and correlations for transparency, but not assigned substantive meaning or used as an explanatory feature.

- A positive Pearson coefficient indicates positive linear association in the analyzed file; it does not establish that intervention on one variable will cause a change in another.
- A small p-value does not indicate the educational importance of an association and does not remove bias caused by unrepresentative sampling.
- An R^2 computed on the same observations used to fit a line is not a test-set accuracy statistic.
- A course-level mean comparison does not establish course quality if cohort composition and assessment difficulty differ or are unknown.
- A dashboard support flag is not a validated risk score until the rule, target outcome, errors, and intervention pathway are documented.
- A generated natural-language summary must be checked against source metrics and reviewed for hallucinations, context loss, and unfairly stigmatizing language.
- A deployed interface is evidence of a software artifact; it is not evidence of improved academic outcomes without a suitable evaluation design.

APPENDIX D
THREATS TO VALIDITY

Construct validity. The meanings of several fields are inferred from their names. The study-hour measurement method, attendance denominator, exam construction, and category-generation rules are not documented. These gaps could materially affect the meaning of the associations. The unknown definition of study_efficiency is why it is not given a substantive explanation.

Internal validity. The analysis is observational and cross-sectional. Prior attainment, motivation, teaching format, course difficulty, external work commitments, health, assessment timing, and many other potential covariates are absent or unverified. The regression lines are therefore descriptive fits, not causal estimates. No intervention was assigned, and no counterfactual comparison was constructed.

External validity. The source, sampling frame, population, institution, collection date, and real-versus-synthetic status are unknown. It is not possible to estimate representativeness or extend the values to other students, regions, institutions, or academic years. The dataset should be treated as a file-level example until provenance is supplied.

Statistical conclusion validity. Pearson correlation captures only linear association, p-values are exploratory, comparisons are not adjusted for multiplicity, and regression assumptions have not been fully tested. Rounding can also make numbers appear slightly different across screenshots and tables. The manuscript reports approximate confidence intervals and provides the full correlation matrix to make the analysis easier to inspect.

Implementation validity. The project artifacts show the visible dashboard screens but do not verify all aspects of the deployed source code, data refresh, access control, report exports, failure handling, mobile behavior, or accessibility. The mismatch between the screenshots and CSV-derived values is direct evidence that the interface needs automated parity

testing before production use.

Conclusion validity. The paper supports conclusions about descriptive patterns in the submitted CSV and the design scope of the current prototype. It does not support claims of predictive validity, reduced workload, increased engagement, fairness, or improved grades. Those are separate empirical questions that require their own data and evaluation designs.

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